Feature Extraction Assignment

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```
# importing libraries
import pandas as pd
import re  # used for text preprocessing
import nltk
from nltk.tokenize import word_tokenize # tokenizing the text data
from nltk.corpus import stopwords
from sklearn.feature_extraction.text import CountVectorizer # converting text to matrix
```

```
# NLTK stopwords data downloading
nltk.download('stopwords')
```

```
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
True
```

```
# Data loading
dataSMS = pd.read_csv(r'/content/SMSSpamCollection.csv', sep='\t', header=None, names=["Class", "SMS_Text"])
```

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```
# datatype of the data
print(dataSMS.dtypes)
```

```
Class       object
SMS_Text    object
dtype: object
```

```
# converting to strings
dataSMS['SMS_Text'] = dataSMS['SMS_Text'].astype(str)
```

```
print(dataSMS.head())
```

```
  Class                                           SMS_Text
0   ham  Go until jurong point, crazy.. Available only ...
1   ham                      Ok lar... Joking wif u oni...
2  spam  Free entry in 2 a wkly comp to win FA Cup fina...
3   ham  U dun say so early hor... U c already then say...
4   ham  Nah I don't think he goes to usf, he lives aro...
```

```
print(dataSMS.head())
```

```
  Class                                           SMS_Text
0   ham  Go until jurong point, crazy.. Available only ...
1   ham                      Ok lar... Joking wif u oni...
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```

```
# Text preprocessing; performing case normalization
dataSMS['PREPROCESSED_SMS'] = dataSMS['SMS_Text'].str.lower()
# removing punctuations/ non word characters and digits
dataSMS['PREPROCESSED_SMS'] = dataSMS['SMS_Text'].apply(lambda x: re.sub(r'\W|\d', ' ', x))
# tokenizing text into words
dataSMS['PREPROCESSED_SMS'] = dataSMS['PREPROCESSED_SMS'].apply(word_tokenize)
```

```
import nltk
nltk.download('punkt')
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Package punkt is already up-to-date!
True
```

```
print(dataSMS.head())
```

```
  Class                                            SMS_Text  \
0   ham  Go until jurong point, crazy.. Available only ...
1   ham                      Ok lar... Joking wif u oni...
2  spam  Free entry in 2 a wkly comp to win FA Cup fina...
3   ham  U dun say so early hor... U c already then say...
4   ham  Nah I don't think he goes to usf, he lives aro...

                                    PREPROCESSED_SMS
0  [Go, until, jurong, point, crazy, Available, o...
1                     [Ok, lar, Joking, wif, u, oni]
2  [Free, entry, in, a, wkly, comp, to, win, FA, ...
3  [U, dun, say, so, early, hor, U, c, already, t...
4  [Nah, I, don, t, think, he, goes, to, usf, he,...
```

```
# remove stopwords
stopwords_english = set(stopwords.words('english'))
dataSMS['PREPROCESSED_SMS'] = dataSMS['PREPROCESSED_SMS'].apply(lambda x: [word for word in x if word not in stopwords_english])
```

```
print(dataSMS.head())
```

```
  Class                                            SMS_Text  \
0   ham  Go until jurong point, crazy.. Available only ...
1   ham                      Ok lar... Joking wif u oni...
2  spam  Free entry in 2 a wkly comp to win FA Cup fina...
3   ham  U dun say so early hor... U c already then say...
4   ham  Nah I don't think he goes to usf, he lives aro...

                                    PREPROCESSED_SMS
0  [Go, until, jurong, point, crazy, Available, o...
1                     [Ok, lar, Joking, wif, u, oni]
2  [Free, entry, in, a, wkly, comp, to, win, FA, ...
3  [U, dun, say, so, early, hor, U, c, already, t...
4  [Nah, I, don, t, think, he, goes, to, usf, he,...
```

```
# converting the list of tokens back to a single string for CountVectorizer
dataSMS['PREPROCESSED_SMS'] = dataSMS['PREPROCESSED_SMS'].apply(lambda x: ' '.join(x))
```

```
print(data_SMS.head())
```

```
  Class                                            SMS_Text  \
0   ham  Go until jurong point, crazy.. Available only ...
1   ham                      Ok lar... Joking wif u oni...
2  spam  Free entry in 2 a wkly comp to win FA Cup fina...
3   ham  U dun say so early hor... U c already then say...
4   ham  Nah I don't think he goes to usf, he lives aro...

                                    PREPROCESSED_SMS
0  Go jurong point crazy Available bugis n great ...
1                          Ok lar Joking wif u oni
2  Free entry wkly comp win FA Cup final tkts st ...
3              U dun say early hor U c already say
4          Nah I think goes usf lives around though
```

```
# creating BoW model
# CountVectorizer instance is created
SMS_vectorizer = CountVectorizer()
# Transformation and Fitting data
SMS_bow = SMS_vectorizer.fit_transform(dataSMS['PREPROCESSED_SMS'])
```

```
# printing out the names of features
print(SMS_vectorizer.get_feature_names_out())
```

```
['____' 'aa' 'aah' ... 'zyada' 'èn' '⩲ud']
```

```
# Transform your BoW matrix into Pandas DataFrame
MS_bow_dfr = pd.DataFrame(SMS_bow.toarray(), columns=SMS_vectorizer.get_feature_names_out())
```

```
print(data_SMS.head())
```

```
  Class                                            SMS_Text  \
0   ham  Go until jurong point, crazy.. Available only ...
1   ham                      Ok lar... Joking wif u oni...
2  spam  Free entry in 2 a wkly comp to win FA Cup fina...
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```

```
                                     PREPROCESSED_SMS
0  Go jurong point crazy Available bugis n great ...
1                           Ok lar Joking wif u oni
2  Free entry wkly comp win FA Cup final tkts st ...
3                U dun say early hor U c already say
4           Nah I think goes usf lives around though
```

```
MS_bow_dfr
```

	____	aa	aah	aaniye	aaooooright	aathi	ab	abbey	abdomen	abeg	...	zhong	zindgi	zoe	zogtorius	zoom	zouk	zs	zyada	èn
0	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0
2	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0
3	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0
4	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
5567	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0
5568	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0
5569	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0
5570	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0
5571	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0	0	0	0

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